

Motivation

Formalization usually comes last:

learn → design → formalize.

This is costly when a protocol depends on several unfamiliar domains.

We reverse the order and treat formalization as a prototyping medium:

idea → formalize → learn.

Inverted Workflow

▶ Human proposes hypotheses, steers abstractions, and sets axiomatization boundaries.

▶ LLM drafts definitions, lemma statements, and candidate proof structure.

▶ Rocq/MathComp checks logical soundness at each step.

▶ Failed obligations reveal missing assumptions early.

Key loop: draft → check → diagnose → refine.

Case Study: SMC-PGG

Secure multi-party computation via covering-space structure:

▶ Monodromy representation $\rho : G \rightarrow S_N$ acts on N sheets.

▶ Secret is encoded in fiber structure over a basepoint.

▶ Session-typed protocol layer implemented in Rocq.

The development is organized as protocol, search-space, reconstruction, and security layers.

Technical Highlights

▶ **Search-space analysis:** RAAG trace equivalence for commuting-action reordering.

▶ **Combinatorics:** Cartier-Foata theorem used for exact trace counting via clique-polynomial recurrence.

▶ **Reconstruction:** threshold interface instantiated through linear-code secret sharing.

▶ **Security bound:**

$$\text{TV} \leq \varepsilon + \frac{2(T - 1)}{N}.$$

▶ **Structure result:** abelian collapse in regular abelian settings; non-abelian structure is required for non-trivial security.

Current Scope and Status

▶ Ongoing formalization: **42 Rocq files**.

▶ Domains touched: group theory, combinatorics, algebraic geometry, coding theory, information theory.

▶ Remaining axioms are concentrated at the algebraic-geometry interface.

Methodological Takeaways

▶ LLMs are useful not only for proof search, but also for cross-domain exploration.

▶ The prover supplies certainty; the human supplies judgment.

▶ Formalization can be an engine for learning, not only final verification.

QR / Links / Contact

Paper draft and artifacts: <https://github.com/<org>/<repo>>

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Replace URL, email, and add QR image if needed.